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PROFESSIONAL SUMMARY

AI/ML Engineer and Associate Technical Team Lead specializing in production-grade cloud-native AI systems, agentic platforms, and LLM-based applications. Expert in Google Cloud Platform (GCP) with hands-on experience in Vertex AI, Cloud Run, Cloud SQL, and Cloud Scheduler. Proven track record leading cross-functional teams in developing multi-agent systems using Agent Development Kit (ADK), implementing A2A (Agent-to-Agent) protocols, and building RAG pipelines for enterprise document analysis. Skilled in FastAPI backend development, MCP (Model Context Protocol) integration, and third-party tool orchestration via Composio. Successfully deployed enterprise solutions including custom Salesforce/HubSpot/ClickUp connectors, Gemini Enterprise setups, intelligent workflow automation platforms, and industry-specific RAG applications for construction and vendor management sectors.

TECHNICAL SKILLS

Cloud & Infrastructure: Google Cloud Platform (GCP), Cloud Run, Vertex AI (Agent Engine, Agent Garden), Cloud SQL, Cloud Scheduler, Cloud Storage, Secret Manager, IAM, VPC Connectors

AI/ML & LLM Technologies: Vertex AI Studio, Gemini API, Gemini Enterprise, Agent Development Kit (ADK), Multi-Agent Systems, RAG (Retrieval-Augmented Generation), Hybrid Search RAG, Vector Embeddings, LangChain, FAISS, Prompt Engineering

Agentic Systems & Protocols: Agent-to-Agent (A2A) Protocol, Model Context Protocol (MCP), Multi-Agent Workflows, Dynamic Prompts, Workflow Templates, Agent Orchestration, Agent Fundamentals

Backend Development: FastAPI, Python, RESTful APIs, Uvicorn, OAuth 2.0, JWT Authentication, Database Design, MySQL Integration

Integration & Orchestration: Composio (3rd Party Tool Integration), Salesforce Bulk API 2.0, HubSpot CRM API, ClickUp API, Google Workspace APIs (Gmail, Drive, Calendar, Docs), Discovery Engine API

Data & ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, XGBoost, catBoost, Hugging Face Transformers

DevOps & Deployment: Docker, Cloud Build, CI/CD, GitHub Actions, Artifact Registry, Streamlit, Gradio, Ngrok

PROFESSIONAL EXPERIENCE

AI/ML Engineer — D3V Technologies (Remote)

June 2025 – Present

Agent Zone – Multi-Agent Agentic Platform

- Architected and developed Agent Zone, a comprehensive multi-agent platform with FastAPI backend, handling complete backend and AI infrastructure independently
- Implemented A2A (Agent-to-Agent) protocol for seamless integration with Salesforce, HubSpot, ClickUp, and Google Workspace, enabling cross-platform agent communication

- Integrated MCP (Model Context Protocol) for all application connectors, utilizing Composio for OAuth authentication flows and third-party tool authorization
- Developed tenant management system, workflow scheduler (manual and scheduled), agentic dynamic prompts, workflow templates, and comprehensive API endpoints
- Built and deployed 7 specialized A2A agents: Orchestrator, HubSpot CRM, Salesforce CRM, ClickUp Task Management, Workspace (Gmail/Drive/Calendar/Docs), Workflow Engine, and Memory Agent

Gemini Enterprise Implementation (Google Product)

- Set up complete Gemini Enterprise infrastructure from scratch for the organization, including data connectors and Discovery Engine configuration
- Implemented ACL (Access Control Lists) for data security, configured authentication and authorization mechanisms, and established third-party connections
- Developed custom Salesforce connector with tri-tier deployment architecture (Interactive Setup, Headless Full Sync, Headless Incremental Sync) using Cloud Run and Cloud Scheduler

Populus – Technical Team Lead

- Led a 3-member technical team as Team Leader, coordinating AI development efforts and project delivery for vendor matching and construction industry solutions
- Developed AI-powered vendor matching system using Gemini API for contact center solution recommendations based on lead attributes
- Integrated SendGrid API for automated personalized email workflows to matched vendors, improving lead follow-up speed by 35%
- Configured environment with Google Cloud Storage for vendor data management and implemented CSV-based data pipeline for vendor-client matching

Populus Sharze Paving Construction Company – RAG-Based Document Analysis System

- Led development of hybrid search RAG-based application as Technical Team Lead for a 3-member team serving construction industry client
- Built intelligent document analysis system to process internal construction documents containing tables, graphs, cost estimations, and technical specifications from Google Drive
- Implemented hybrid RAG architecture combining vector embeddings and traditional search to enable accurate document summarization and construction-specific query responses
- Enabled automated extraction and analysis of project estimates, material costs, and construction timelines from complex multi-format documents
- Tech Stack: Python, Vertex AI, RAG, BigQuery Vector Search, Google Drive API, Document AI, Gemini API, FastAPI

Vertex AI RAG Agent

- Built production-ready RAG agent using Vertex AI and Google Drive integration with automatic document classification and preprocessing
- Implemented corpus management system for creating, querying, and managing document collections with Gemini-powered preprocessing for LAB_REPORT and ODOT_DOCUMENT files
- Deployed agent with ADK (Agent Development Kit) web interface, enabling document ingestion, RAG queries, and corpus operations

Additional Key Contributions:

- Built OLLI GCP Billing Assistant using Gemini Pro API with <1.2s latency and >90% response accuracy for real-time billing queries

- Contributed to Google Agent Entrepreneur internal automation using Vertex AI, Dialogflow CX, and Sheets API for CRM operations

AI & ML Intern — VRT Management Groups LLC, Hyderabad, India

January 2025 – June 2025

- Engineered AI-powered attendance system using OpenCV and Gradio, reducing processing time by 30%
- Fine-tuned OpenAI GPT model for NLP automation, achieving 90% accuracy in text generation tasks
- Developed Streamlit app for Ideal Customer Profile analysis, improving segmentation accuracy by 15%
- Deployed machine learning models via FastAPI with optimized feature engineering and hyperparameter tuning

Data Science Intern — Coincent, Bangalore, India

April 2024 – June 2024

- Built salary prediction model using XGBoost, achieving 83% accuracy and enhancing workforce planning
- Integrated MySQL with Streamlit for real-time dashboards, improving data visualization efficiency by 25%

ACADEMIC PROJECTS

WhatsApp AI Conversational Chatbot / February 2024 – April 2025

- Developed intelligent chatbot integrated with WhatsApp using Gemini 2.0 Flash model, deployed via Flask and Twilio WhatsApp API
- Implemented real-time message handling with Ngrok for local tunneling and secure API key management
- Enabled features like casual dialogue and appointment booking with <1-second latency. Tech Stack: Python, Flask, Twilio, Gemini API, NLP

RAG Model Development for Knowledge Retrieval / November 2025

- Designed RAG model using LangChain and Hugging Face Transformers for context-aware responses in knowledge retrieval system
- Implemented vector embeddings with FAISS for efficient document retrieval, reducing response time by 25%
- Fine-tuned model with 5,000-document dataset, achieving 88% accuracy. Tech Stack: Python, LangChain, Hugging Face, FAISS, spaCy, NLP

ICP Prediction and Lead Filtering System / January 2024 – February 2025

- Built ML-based system to prioritize Ideal Customer Profiles and automate lead filtering for B2B sales pipeline
- Utilized Scikit-learn for KNN and decision tree models, achieving 15% improvement in lead conversion rates
- Developed Streamlit dashboard for real-time visualization, reducing manual effort by 30%. Tech Stack: Python, Scikit-learn, Streamlit, Pandas, SQL

EDUCATION

B.Tech in Computer Science Engineering

KMM Institute of Technology and Science (Affiliated with JNTUA)

2021 – 2025 | Academic Score: 86%

CERTIFICATIONS & SKILL BADGES

Google Cloud Skill Badges:

- Introduction to Vertex AI Studio
- Agent Fundamentals
- Generative AI for Business Leaders
- Create Embeddings, Vector Search, and RAG with BigQuery
- Plan Change Management for Gemini Enterprise Deployments

Professional Certifications:

- Introduction to Data Science – Cisco (2024)
- Python – HackerRank (2024)
- SQL – HackerRank (2024)

LEADERSHIP & ACHIEVEMENTS

- Technical Team Leader for Populus AI projects, leading a 3-member team in developing enterprise AI solutions for vendor matching and construction industry document analysis
- Awarded Pratibha Award 2025 by Government of Andhra Pradesh for outstanding academic performance and state-level paper presentation on Business Development and AI
- Led team to 1st place in KSRM Kadapa ML/AI Competition 2023, achieving 89% model accuracy and outperforming 50+ teams
- Received 6 consecutive Academic Excellence Awards (Semesters 2-1 to 4-2) for maintaining top 1% rank at KMM Institute
- Captained Chittoor District Hockey Team to regional championship victories, demonstrating leadership and team coordination
- Served as Class Representative (CR) for 3+ semesters, facilitating communication between students and faculty